

Scalability & Future Plan

Date	28 September 2026
Team ID	SWTID-2026-6259
Project Name	AI StudyBuddy API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.N o	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited scalability for a large number of concurrent users	May affect response time and system performance	High
2	Large amounts of study materials and AI-generated resources may increase database load	May affect storage and data retrieval performance	Medium
3	Dependence on the Google Gemini API for AI-generated content	API availability and usage limits may affect AI features	Medium

Scalability Plan:

S.N o	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Designed for normal student usage	Use cloud deployment and load balancing to support more concurrent users
2	Data Storage	MongoDB stores user data, study materials, and AI resources	Scale MongoDB storage and optimize database queries and indexing

3	Performance	REST APIs process requests through the backend	Use caching and optimized API processing to improve response time
4	Security	JWT authentication, bcrypt password encryption, and RBAC	Strengthen authentication, authorization, and secure API access

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Provides easier access to learning resources		
Phase 3	Provides more personalized learning experiences		
Phase 4	Supports larger user bases and future platform expansion		